TypeScript Introduction (Detailed Guide for PDF)

What is TypeScript?

TypeScript is a superset of JavaScript that adds static typing and advanced features to help build large-scale applications.

History of TypeScript

- Developed by Microsoft
- Released in 2012
- Designed by Anders Hejlsberg

Why Use TypeScript?

- Detect errors early (compile-time)
- Better code readability
- Scalable for large applications
- Strong tooling support

Features of TypeScript

- Static typing
- Interfaces
- Classes
- Modules
- ES6+ features support

Installing TypeScript

```
npm install -g typescript
```

Compiling TypeScript

```
tsc file.ts
```

Basic Syntax

```
let message: string = "Hello TypeScript";
console.log(message);
```

Data Types

- number
- string
- boolean
- any
- void
- null
- undefined

Example:

```
let age: number = 25;
let name: string = "John";
```

Functions

```
function add(a: number, b: number): number {
  return a + b;
}
```

Interfaces

```
interface Person {
  name: string;
  age: number;
}
```

Classes

```
class Student {
  name: string;
```

```
  constructor(name: string) {
    this.name = name;
  }
}
```

Arrays

```
let numbers: number[] = [1, 2, 3];
```

Tuples

```
let person: [string, number] = ["John", 25];
```

Enums

```
enum Color { Red, Green, Blue }
```

Type Assertion

```
let value: any = "Hello";
let length: number = (value as string).length;
```

Modules

```
export function greet() {
  console.log("Hello");
}
```

TypeScript with React

Used to build scalable frontend apps.

Example Program

```
function greet(name: string): string {
  return "Hello " + name;
}
```

```
console.log(greet("John"));
```

Advantages of TypeScript

- Early error detection
- Better maintainability
- Strong typing system

Disadvantages of TypeScript

- Compilation step required
- Slight learning curve

Best Practices

- Use strict typing
- Avoid using 'any'
- Organize code into modules